

GROUP 2

CREDIT CARD FRAUD

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STAGE 0: PROBLEM, DATA AUDIT, CLEANING & EDA

Task: Classify each transaction as legitimate (Class 0) or fraudulent (Class 1). Missed fraud costs banks and customers real money.

Dataset: 284807 real transactions, 31 columns (V1 - V28 are anonymised PCA features)

Missing Values: Audit found 0 missing values across all 31 columns - no imputation needed

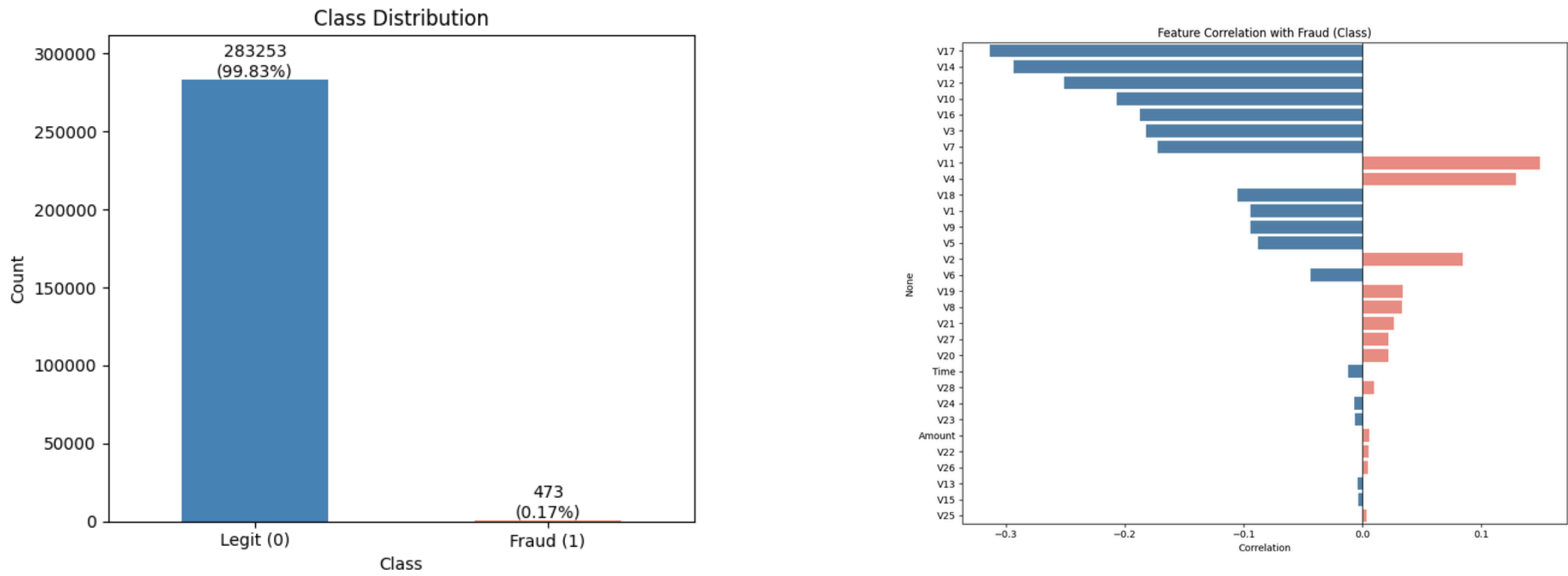
Duplicates: 1081 duplicate rows removed (284807 → 283726) to prevent biased training

Outliers: Detected with IQR (highest: V27 with 13.67, Amount with 11.17%) - in fraud detection, the extreme values are a signal; RobustScaler later limits their influence.

EDA Insight 1: Extreme imbalance. 283253 legit vs 473 fraud. “Always legit” scores 99.83% accuracy yet catches zero fraud. Thus, accuracy is the wrong metric.

EDA Insight 2: V17, V14, V12, V10, and V16 correlate most with fraud, but no single feature separates the classes. The model must combine features (3-model comparison).

Feature engineering: Time → cyclical Hour_sin / Hour_cos (hour 23 and hour 0 become neighbours); Amount → log1p, skewness 16.98 → 0.16.



STAGE 0b: HANDLING THE 0.17% CLASS IMBALANCE

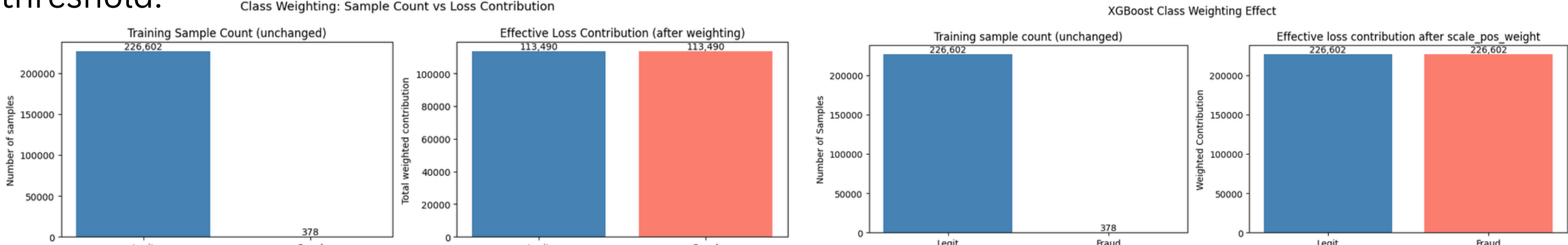
Split first, handle imbalance after: Stratified 80/20 split (226,980 train / 56,746 test);

RobustScaler and all imbalance handling are fitted on training data ONLY – the test set is never touched (no leakage).

Methods compared on Logistic Regression: Unweighted vs class weighting vs SMOTE. Class weighting and SMOTE both lifted fraud recall from 0.59 to 0.87; SMOTE offered no advantage over weighting (AUC-PR 0.6779 vs 0.6807), so the simpler class weighting was adopted across the pipeline.

The weights: legit 0.5008 vs fraud 300.2381 – one fraud example contributes as much loss as ~600 legit examples, forcing every model to pay attention to the rare class without changing the data.

Metric choice (why AUC-PR): with 0.17% fraud, accuracy is meaningless and ROC-AUC is inflated by the huge legitimate class. AUC-PR summarises the precision–recall trade-off across all thresholds – our primary metric; recall, precision and F1 for the fraud class reported at the 0.50 threshold.



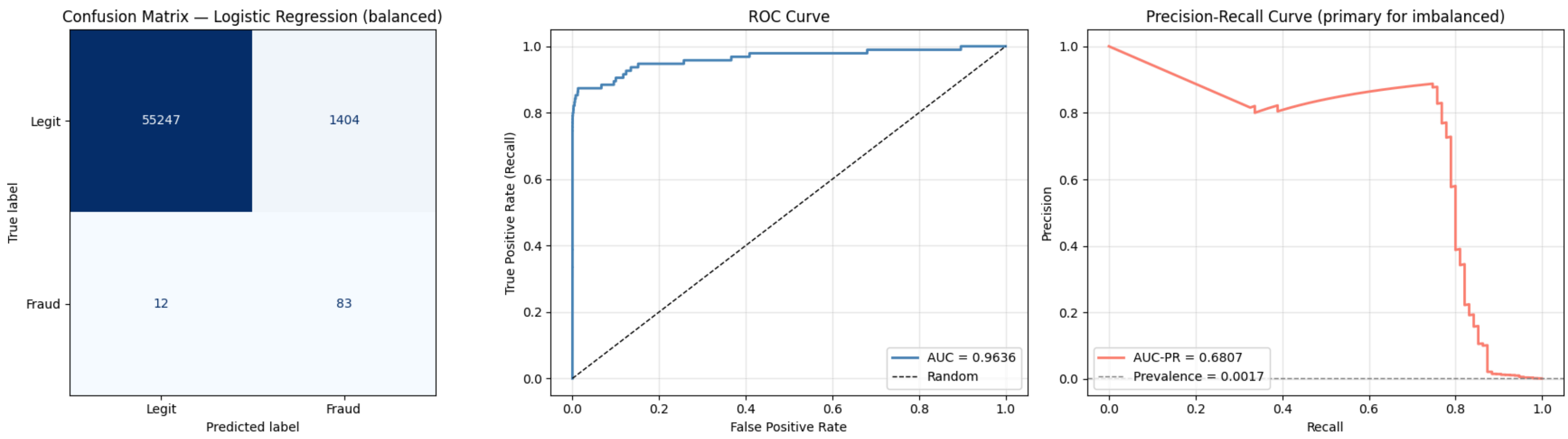
STAGE A: LOGISTIC REGRESSION

How it works: learns one weight per feature, sums them, and squashes the result through a sigmoid into a fraud probability – a single straight boundary through feature space.

Setup: class_weight='balanced', lbfgs, max_iter=1000, on the 31 RobustScaler-scaled features. Strongest coefficients: V4 (+1.66), V12 (−1.37), V14 (−1.28), V10 (−1.27) – matching the EDA.

Result: recall 0.8737 (83 of 95 frauds caught) but precision only 0.0558 – 1,404 false alarms; AUC-PR 0.6807.

Why it fails: A single straight line cannot tightly enclose the fraud region – pushing recall up inevitably sweeps in legitimate transactions. Useful as a fair reference point; not deployable.



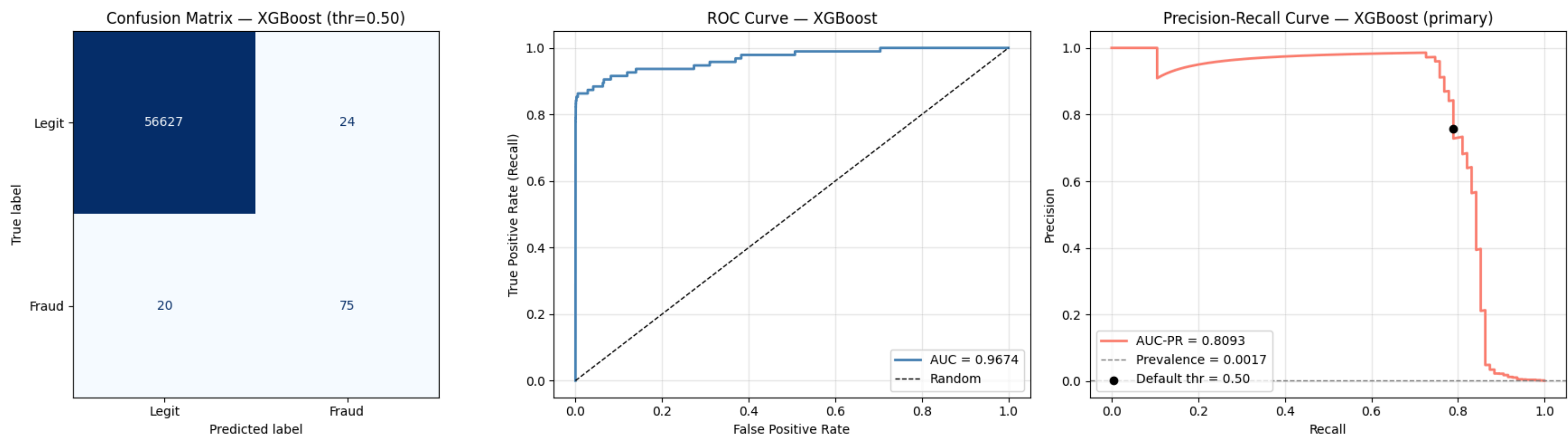
STAGE B - XGBoost

Setup: scale_pos_weight = 599.48 (legit : fraud ratio). RandomizedSearchCV, 10 configs × 5-fold stratified CV (50 fits), scored by AUC-PR. Best: 200 trees, depth 4, learning rate 0.1

Result: AUC-PR 0.8093 (best), precision 0.7576, recall 0.7895, F1 0.7732 – 75 frauds caught, 20 missed, only 24 false alarms vs the baseline’s 1,404.

Weighting Effect: scale_pos_weight lifted AUC-PR 0.7940 → 0.8093 and recall 0.7158 → 0.7895, trading precision (0.9577 → 0.7576), the right trade when missed fraud is the costly error

Key Features: V14 ≈ 35% importance, then V12 & V10 ≈ 10% importance; same features flagged by EDA

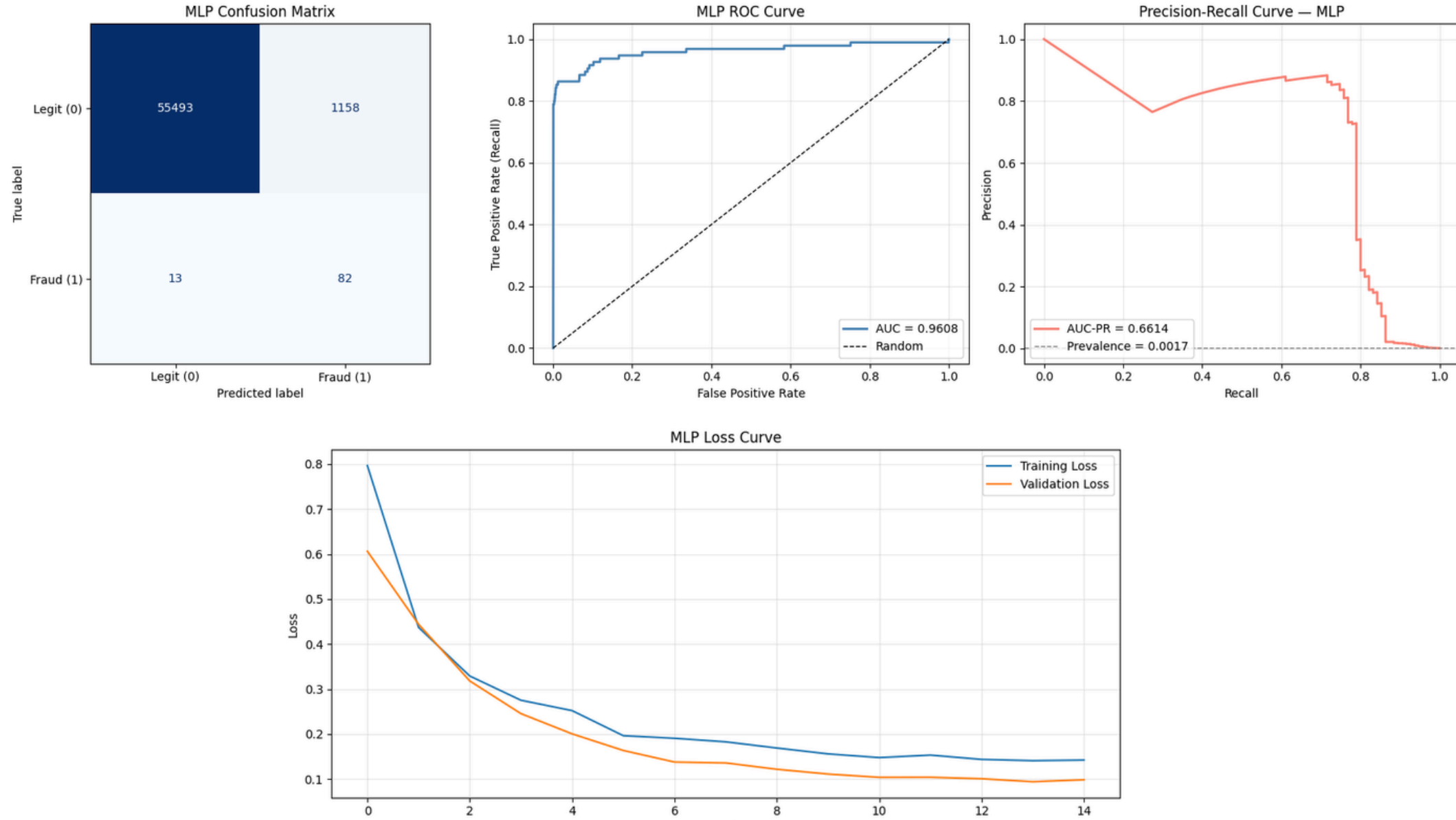


STAGE C: DEEP LEARNING

Setup: 31 → Dense(32, ReLU) → Dropout 0.2 → Dense(16, ReLU) → Dropout 0.2 → Dense(1, sigmoid); 1,569 parameters; Adam; class-weighted binary cross-entropy; 15 epochs, batch 2048.

Result: AUC-PR 0.6614 – below even the baseline. Recall 0.8632 (82 frauds) but precision 0.0661 with 1,158 false alarms.

Why: Loss curves show healthy training (no overfitting), the network simply found no better representation. With only 378 training frauds and features that are already PCA components, a small tabular dataset is exactly where tree ensembles beat small neural networks.



FINAL COMPARISON: STAGE A vs B vs C

Model	Precision	Recall	F1	AUC-PR
Logistic Regression	0.0558	0.8737	0.1049	0.6807
XGBoost	0.7576	0.7895	0.7732	0.8093
MLP	0.0661	0.8632	0.1228	0.6614

